Explicit Modular Periods for Ordered Partitions into Odd Blocks

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Abstract

The number of ordered set partitions into odd blocks is eventually periodic modulo every positive integer. We prove the conjecture in OEIS A006154 with an explicit period dividing twice the Euler totient of the modulus; for odd moduli the factor two is unnecessary. A differential-operator congruence treats all prime powers uniformly. The least eventual period modulo ten is four, beginning at index one.

Let $a(n)$ count ordered partitions of an n-element set into blocks of odd cardinality, with $a(0) = 1$. Choosing the first block gives

$$a(n) = \sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} a(n-j), \qquad F(x) = \sum_{n \ge 0} a(n) \frac{x^n}{n!} = \frac{1}{1 - \sinh x}.$$

OEIS A006154 records this sequence and Peter Bala's April 12, 2023 conjecture that its reduction modulo every positive integer is eventually periodic [1].

Theorem 1. *Factor $m \ge 2$ into prime powers. Define*

$$P(m) = \operatorname{lcm}\left(\{\varphi(p^e) : p^e \parallel m,\ p \text{ odd}\} \cup \{2^e : 2^e \parallel m\}\right).$$

The second set is empty for odd m. For every $n \ge m$,

$$a(n + P(m)) \equiv a(n) \pmod{m}.$$

In particular $P(m) \mid 2\varphi(m)$; for odd m, $P(m) \mid \varphi(m)$. Modulo one the sequence is constant.

1 Differential congruences

Work in the torsion-free ring $A = \mathbb{Z}[s, c, (1-s)^{-1}]$, and put $u = (1-s)^{-1}$. The derivation

$$D = c\partial_s + s\partial_c$$

preserves A, since $D(u) = cu^2$. Substitution $(s, c) = (\sinh x, \cosh x)$ turns D into differentiation with respect to x. Thus, with $\varepsilon(f) = f(0, 1)$,

$$a(n) = \varepsilon(D^n u). \tag{1}$$

This formula uses only integer expressions and no division by factorials modulo a modulus.

Lemma 1. *As endomorphisms of A,*

$$D^p \equiv D \pmod{p} \quad (p \text{ odd prime}), \qquad D^4 \equiv D^2 \pmod{2}.$$

Proof. In characteristic p, the pth iterate of a derivation is again a derivation, by the iterated product rule and the divisibility of $\binom{p}{j}$ for $0 < j < p$. For odd p, the derivations D^p and D agree on s, c, since $Ds = c$ and $Dc = s$. They therefore agree on the polynomial ring and its localization. In characteristic two both D^4 and D^2 are derivations; on each of s, c they equal the identity. They agree on the localization as well. This proves both operator congruences. □

Lemma 2. *Let X, Y be commuting integer-linear endomorphisms of a torsion-free abelian group. If $X \equiv Y \pmod{p^r}$, $r \geq 1$, then $X^p \equiv Y^p \pmod{p^{r+1}}$.*

Proof. Write $X = Y + p^r Z$. The endomorphism Z is integer-linear and commutes with Y, by torsion-freeness. In the binomial expansion of $(Y + p^r Z)^p - Y^p$, the first term has factor p^{r+1}. For $2 \leq j < p$ the jth term has factor p^{rj+1}, and the last has factor p^{rp}, where $rp \geq r + 1$. Every term is divisible by p^{r+1}. □

Proof of Theorem 1. Repeatedly apply Lemma 2 to the commuting powers of D in Lemma 1. This gives

$$D^{p^e} \equiv D^{p^{e-1}} \pmod{p^e} \quad (p \text{ odd}), \qquad D^{2^{e+1}} \equiv D^{2^e} \pmod{2^e}.$$

Multiplication by further powers of D and (1) prove periods $p^{e-1}(p-1)$ from index p^{e-1} for odd p, and period 2^e from index 2^e for the power of two. Every threshold is at most m. The Chinese remainder theorem gives their least common multiple as a period modulo m. Finally each odd-prime period is the corresponding factor of $\varphi(m)$, and the power-of-two period is twice that factor, proving the divisibility assertions. □

2 The exact period modulo ten

Modulo five, Lemma 1 gives period four from index one. To handle the initial indices modulo two, reduce in

$$V_2 = (\mathbb{Z}/2\mathbb{Z})[s, c]/(s^2, c^2 - 1).$$

The derivation preserves the ideal and $u = 1 + s$ in this quotient. Consequently

$$Du = c, \qquad D^2 u = s, \qquad D^3 u = c = Du.$$

Evaluation at $(0, 1)$ descends to V_2, proving period two from index one. Combining these results gives

$$a(n) \bmod 10 = (1, 2, 7, 2)_{\,1+((n-1) \bmod 4)} \qquad (n \geq 1).$$

The block has neither period one nor period two, so its least eventual period is four. This also proves the specific example recorded with the general conjecture.

The supplementary verifier compares all 201 published terms with the first-block recurrence and independently enumerates literal set partitions through eight elements. Finite differential quotients check the operator calculation for moduli two through 32. The proof above establishes the universal periodicity; the finite checks audit its implementation.

References

[1] The OEIS Foundation, A006154, including P. Bala's periodicity conjecture (2023) and differential formula (2011), https://oeis.org/A006154, accessed October 27, 2026.